
Spectral ECG Analytics: AI-Driven Early Detection of Sudden Cardiac Arrest

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Abstract: A cardiovascular condition known as Sudden Cardiac Arrest (SCA) strikes a person so quickly that it requires the least amount of inpatient time and may even result in death. Rapid and very accurate identification of SCA shortly after the onset of symptoms is essential to the patient's survival. A standardized technique for SCA identification with high accuracy and in a short amount of time is still required, despite the fact that many approaches based on electrocardiograms have been used. Therefore, the current study suggests using four intelligent classifiers based on temporal data taken from ECG to detect SCA. Performance evaluation of the proposed classifiers on 185 SCA and 182 normal subjects, yield classification accuracies of 95.6%, 90.46%, 93.73%, 95.91% for Decision Tree, SVM, KNN and Neural Network respectively.

Keywords: Electrocardiogram (ECG), Sudden Cardiac Arrest (SCA), Spectral Features, Welch Power Spectral Density (PSD), Machine Learning, Biomedical Signal Processing, Classification

INTRODUCTION

When the heart abruptly stops due to abnormal electrical activity—most commonly ventricular fibrillation—it results in sudden cardiac arrest (SCA), a life-threatening condition requiring immediate medical intervention. Electrocardiogram (ECG) signals provide critical insight into cardiac activity, comprising key components such as the P-wave (atrial depolarization), QRS complex (ventricular depolarization), and T-wave (ventricular repolarization). In healthy individuals, intervals such as Q–T and R–R exhibit regular periodicity, whereas deviations from these patterns often indicate underlying cardiac abnormalities.

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Among the potential reasons include congenital cardiac defects, heart attacks, cardiomyopathy, coronary artery disease, and electrolyte imbalances. Sudden and severe symptoms, such as collapse, unconsciousness, lack of breathing, and pulse, can occur. Cardiopulmonary resuscitation (CPR), using an automated external defibrillator, and contacting emergency services are all necessary for immediate care. Implanted cardioverter-defibrillators (ICDs), medications, and therapy for underlying cardiac issues may be part of long-term care. With more than 300,000 deaths recorded each year in the US alone, SCA is a serious global health concern [2]. In India, SCA claims the lives of about 4280 out of 1 lakh persons annually [3]. Before a sudden cardiac arrest, palpitations, weakness, chest tightness, shortness of breath, and rapid or irregular heartbeats are common symptoms. Despite many studies investigating the use of ECG features for SCA diagnosis, a technique that can quickly and accurately identify SCA is still required. Rashed et al. conducted a study employing spectral domain analysis of electrocardiograms to demonstrate the detection of sudden cardiac death [4]. Through the analysis of ECG repolarization irregularities, Nicolas et al. identified a technique for diagnosing Sudden Cardiac Arrest (SCA) [5]. Kim et al. [6] proposed a strategy for recognizing cardiac arrest in octogenarians and nonagenarians in non-hospital settings. Vijaya et al. outlined a method utilizing the Pan-Tomkins algorithm to detect sudden cardiac arrest [7]. A study employed a modified wavelet transform to detect SCA [9], whereas Murukesan et al. offered a method for diagnosing SCA through heart rate variability utilizing a machine learning technique with improved accuracy [8]. Shen et al. claimed 75% accuracy with their HRV-FFT neural network model [13], Parsi et al. claimed 90.2% accuracy with HRV-based features [12], Rohila et al. claimed 91.67% accuracy with SVM and Decision Trees [11], and Devi et al. claimed 83.33% accuracy with HRV and wavelet features [10]. This study examines four machine learning classifiers that use the Physio net ATM database and ECG-derived characteristics to detect SCA. The QRS complex and T-wave were then recognized using a modified Pan Tomkins algorithm, and the RR and QT intervals were calculated from the discovered peaks. However, spectral properties that are taken from the frequency domain give ECG signals a more reliable representation. Ratio values like spectral energy ratio, centroid ratio, maximum power ratio, and band power ratio (θ/α) have been identified, which help detect the variations in autonomic nervous system activity as well as the hidden periodicity that is not easily detectable using time series data. The efficiency of the proposed technique will enhance the accuracy and reliability of machine learning classifiers used to distinguish between healthy and SCA subjects. In particular, the framework has shown great potential with respect to the development of hardware based on the very large scale integration methodology.

While traditional methods mainly depend on either the use of single-domain attributes or individual classifier systems, this research involves the incorporation of efficient spectral features extraction and an evaluation process based on the comparison of several machine learning algorithms.

This work's contributions:

- Creation of a preprocessing pipeline for ECG denoising that uses bandpass filtering.
- The 2 Hours signal extracted from Physionet ATM database is divided into 4

- segments each of 30 mins
- Extraction of more noise-resistant frequency-domain spectrum features (energy, centroid, max power, and bandpower ratios) as opposed to time-domain features were performed each on 30 minutes segmented signal.
 - Comparative analysis of several classifiers for SCA identification, including SVM, Decision Tree, KNN, and Neural Network.
 - An end-to-end technique block diagram in its entirety.

BLOCK DIAGRAM

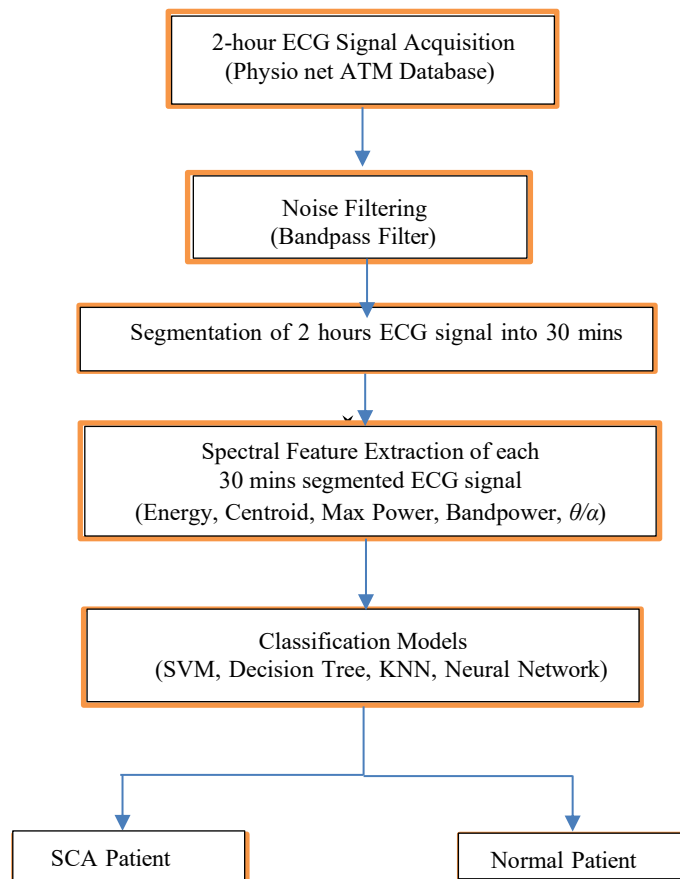


Figure 1: Block Diagram

METHODOLOGY

The suggested approach for employing machine learning and ECG signals to identify Sudden Cardiac Arrest (SCA) is displayed in Fig. 1. The workflow consists of six main stages:

A. ECG Signal Acquisition:

The Physio net ATM database, which is frequently utilized in cardiac research, provided the ECG data. In order to provide sufficient temporal resolution, the signals were sampled at 250 Hz.

B. Preprocessing:

Motion distortions, baseline drift, and noise were removed from the ECG signals using a bandpass filter. This method preserves only therapeutically meaningful frequency components.

C. Segmentation:

2-hour ECG Signals was taken and the Signals were segmented into 4 parts each containing 30 mins ECG Signal. The feature extraction was performed on these each 4 segmented 30 mins ECG Signal.

D. Feature Extraction:

Welch's approach was used to extract spectral characteristics from the frequency domain rather than just time-domain features. The following were calculated:

- Total Spectral Energy
- Spectral Centroid
- Frequency with Maximum Power
- Total Band power (Delta, Theta, Alpha, Beta bands)
- Theta-Alpha Power Ratio

A stronger depiction of the underlying cardiac dynamics is offered by these characteristics.

E. Classification:

The retrieved spectral characteristics were subjected to four machine learning models:

- Support Vector Machine (SVM)
- Decision Tree
- K-Nearest Neighbors (KNN)
- Neural Network (NN)

1) Support Vector Machine (SVM): One kind of classification method that divides data points using a hyperplane is the support vector machine. Support vectors are certain data points used to build the hyperplane. Despite being a linear classifier, SVM can do non-linear classification by employing a kernel method [15]. The RBF kernel is employed in this suggested approach. To distinguish between SCA and normal classes, SVM creates an ideal hyperplane in the feature space. In this study, spectral characteristics were used to train SVM for binary classification.

2) Decision Tree (DT): The tree model in decision trees is created by the use of thresholds in order to segment the feature space. This means that the Decision Tree classifier creates the tree structure through the use of conditional parameters and the parameter values, which are determined from the training data. After the optimal parameters are used to build the tree structure, classification amongst the classes is done using the decision tree algorithm.

3) K-Nearest Neighbors (KNN): The K-Nearest Neighbour algorithm is a straightforward method for classification that uses the calculation of the Euclidean distance between the test input and a cluster of points from the training set [18,19]. The output class of the test point is determined through a user defined constant value, K.

4) Neural Network (NN): The training process of the neural network utilized spectral characteristics. As the neural network employs hidden layers with non-linear activation functions, it can effectively learn intricate feature-class relationships to predict SCA accurately.

To identify the top-performing classifier, the models were trained and assessed on labeled ECG segments (normal vs. SCA).

F. Output Decision:

The classification models produced two potential results:

- SCA Patient: It suggests a risk of unexpected cardiac arrest.
- Non SCA Patient (Normal): shows a normal heart rhythm. The classification allows for early evaluation and treatment for patients.

EVALUATION PERFORMANCE METRICS

To check the performance of the proposed system, standards such as accuracy, sensitivity, specificity and precision were used. These actions helps in checking the reliability of the classification models in detecting SCA (Sudden Cardiac Arrest).

A. Accuracy:

Accuracy refers to the ratio of instances that are correctly classified compared to the total number of samples. It is described as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (1)$$

B. Sensitivity (Recall/True Positive Rate):

Sensitivity tells us how well the SCA cases are identified by the model. It is given by:

$$\text{Sensitivity} = \frac{TP}{TP+FN} \quad (2)$$

C. Specificity (True Negative Rate):

The specificity of the model says how well it can detect non-SCA (normal) patients. It is explained as:

$$\text{Specificity} = \frac{TN}{TN+FP} \quad (3)$$

D. Precision (Positive Predictive Value):

Precision determines the percentage of anticipated SCA situations that turn out to be accurate. It is said as follows:

$$\text{Precision} = \frac{TP}{TP+FP} \quad (4)$$

Here, TP = True Positives, TN = True Negatives, FP = False Positives, FN = False Negatives.

RESULTS AND DISCUSSION

- A. Feature Authentication: It is clear from the Physio net ATM Database study that there is a marked difference between the spectral characteristics of the ECG signal in healthy individuals and those prone to Sudden Cardiac Arrest (SCA). The higher values of centroid in healthy individuals point out that there is balanced energy distribution among the frequencies while the centroid of SCA individuals shows lower values, which means that the energy of the ECG in SCA individuals is shifted to lower frequencies. Likewise, the band power study reveals that the energy concentration of θ frequency range in SCA individuals is higher than that of healthy individuals.

These results show that spectral-domain features such as energy, band power, and centroid serve as effective differentiators for distinguishing SCA from normal conditions. They provide dependable input parameters for additional machine learning models aimed at accurately and promptly identifying SCA.

- B. Performance Analysis: Both scatter plot of expected versus actual labels and Receiver Operating Characteristics (ROC) curves are checked by the classification models' performance. It shows the accuracy of each model to

distinguish patients for Sudden Cardiac Arrest (SCA).

1. ROC Curve Analysis: Figure 2 shows the ROC Curve for the Decision Tree Classifier. Area Under the Curve (AUC) demonstrates the model's ability to distinguish SCA patient from healthy persons.

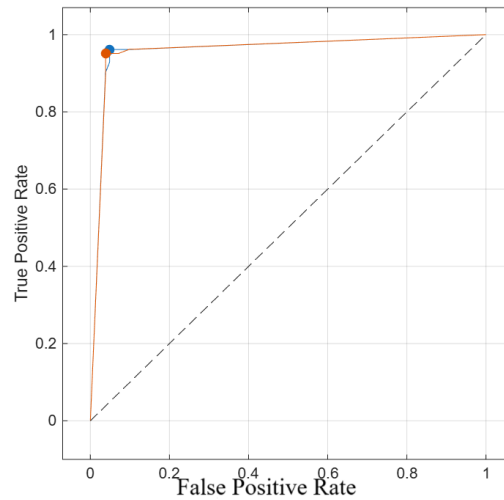


Figure 2: ROC Curve of Decision Tree classifier.

2. ROC Curve Analysis: Figure 3 shows the ROC Curve for the SVM Classifier. Area Under the Curve (AUC) demonstrates the model's ability to distinguish SCA patient from healthy persons.

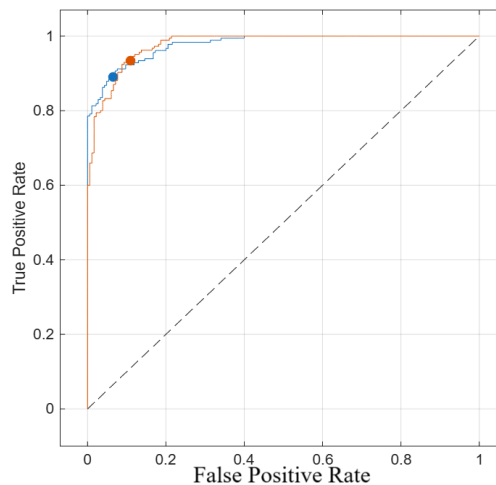


Figure 3: ROC Curve of Support Vector Machine (SVM)

3. ROC Curve Analysis: Figure 4 shows the ROC Curve for the KNN Classifier. Area Under the Curve (AUC) demonstrates the model's ability to distinguish SCA patient from healthy persons.

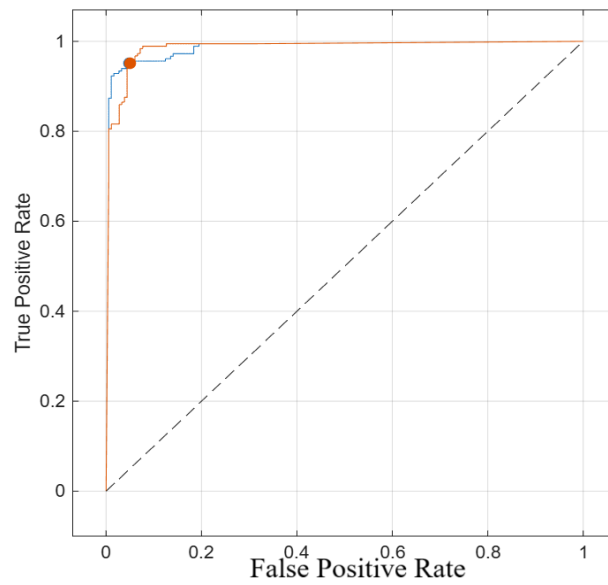


Figure 4: ROC Curve of KNN (K-Nearest Neighbors).

4. ROC Curve Analysis: Figure 5 shows the ROC Curve for the Neural Network Classifier. Area Under the Curve (AUC) demonstrates the model's ability to distinguish SCA patient from healthy persons.

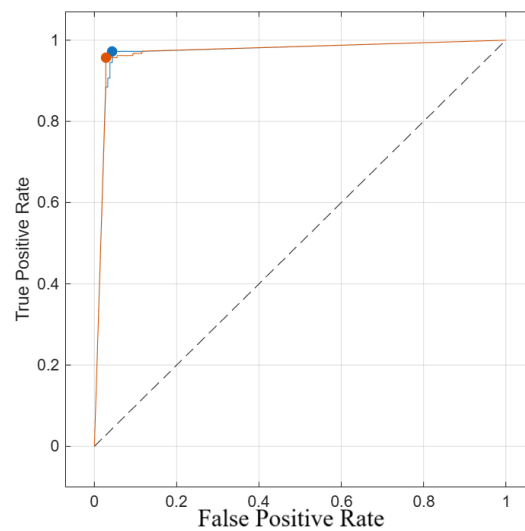


Figure 5: ROC Curve of Neural Network Classifier.

5. Scatter Plot Analysis: Decision Tree: Figure 6 shows the scatter plot for the Decision Tree Classifier. The divergence between expected and actual classifications is illustrated by this scatter plot.

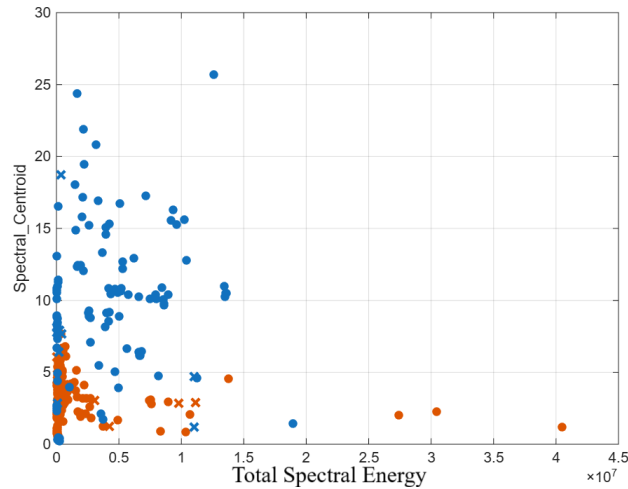


Figure 6: Scatter plot of Decision Tree

6. Scatter Plot Analysis: Support Vector Machine (SVM): Figure 7 shows the scatter plot for the SVM. The divergence between expected and actual classifications is illustrated by this scatter plot.

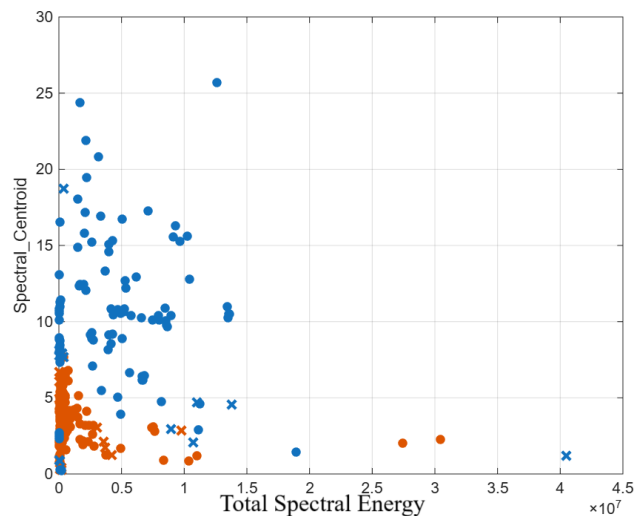


Figure 7: SVM classifier scatter plot of expected versus actual labels.

7. Scatter Plot Analysis: K-Nearest Neighbors (KNN): Figure 8 shows the scatter plot for the KNN. The divergence between expected and actual classifications

is illustrated by this scatter plot.

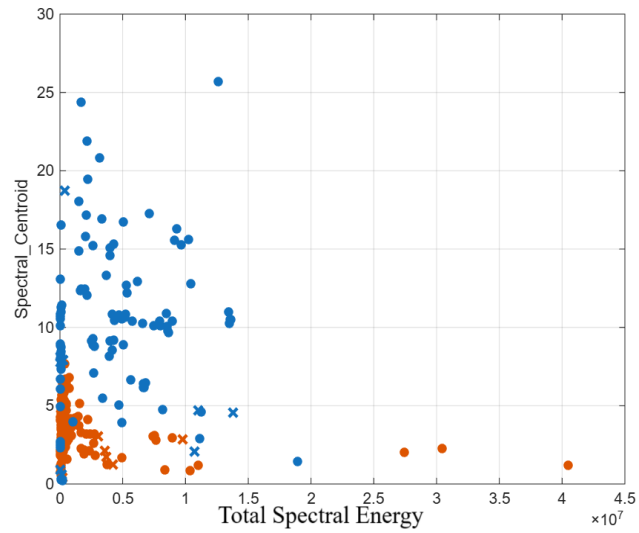


Figure 8: Scatter Plot of K-Nearest Neighbors

8. Scatter Plot Analysis: Neural Network: Figure 9 shows the scatter plot for the Neural Network. The divergence between expected and actual classifications is illustrated by this scatter plot.

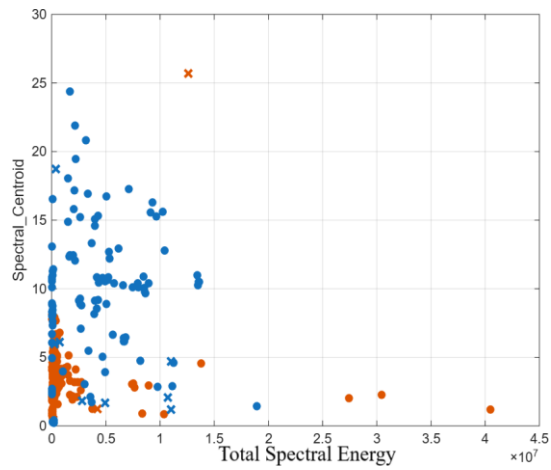


Figure 9: Scatter Plot of Neural Network

9. Heatmap Analysis: In Figure 10, the heatmap shows the relationship between the retrieved spectral features and the model predictions. More powerful associations are represented by more intense or brighter colors. This chart will help us determine which factors are essential in distinguishing between SCA patients and healthy individuals.

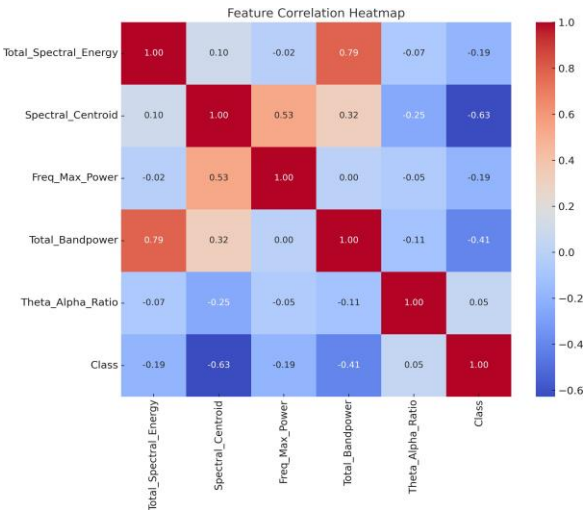


Figure 10: A heatmap that displays the relationship between model predictions and spectral characteristics.

10. Performance Metrics: Table 1 shows the performance of the classifiers in detecting the SCA

TABLE 1: Performance metrics of classification models for SCA detection (%)

Model	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)
Decision Tree	95.64	95.14	96.15	96.17
SVM	90.46	92.97	87.91	88.66
KNN	93.73	94.05	93.41	93.55
Neural Network	95.91	95.68	96.15	96.20

11. Comparative Study: In conjunction with the existing study based on the spectral features, Table 2 shows a comparative analysis of the previously established approaches for detecting SCA.

TABLE 2: Comparative study of the present work with previously reported methods (Accuracy is in %)

Reference (Description)	Accuracy (%)
Devi et al., 2023 (wavelet features)	83.33 %
Rohila et al., 2023 (SVM and Decision Trees using HRV)	91.67 %
Parsi et al., 2023 (HRV-based features)	90.2 %
Shen et al., 2023 (HRV-FFT Neural Network model)	75 %
Present Work (Spectral features from ECG signals)	95.91 %

CONCLUDING REMARKS

The paper has proposed a novel scheme for the early diagnosis of Sudden Cardiac Arrest (SCA) through the employment of spectral characteristics extracted from ECG signals. The experiment involved the use of a data set consisting of 185 SCA patients and 182 normal individuals for validating four classification techniques – Support Vector Machine (SVM), Decision Tree, K-Nearest Neighbors (KNN), and Neural Network.

The results indicated that spectral domain variables such as total energy, spectral centroid, peak frequency, band power, and theta-alpha power ratio have high discriminative ability in separating patients with SCA from control subjects. The best accuracy was achieved by both the Neural Network model and Support Vector Machine classifier compared with the prior literature, demonstrating the effectiveness of spectral variables over time-domain and HRV methods. The robustness of the proposed models was evaluated using ROC curves, scatter plots, and heatmap analysis. According to the analysis, the achieved accuracy was 95.91%.

Limitations: Although it shows promising outcomes, this study had few drawbacks:

- The dataset was small and was taken from Physionet ATM; larger and more varied datasets are required for greater generalization.
- Pre-recorded ECG data was used to develop these models;
- Only spectral characteristics were taken into account; integrating time-domain, frequency-domain, and nonlinear characteristics could enhance performance even more.

Concluding, for detecting SCA at an early stage, this paper provides a reliable, efficient, and comprehensible model, despite the fact that additional effort is required for its real-world application. In addition, the proposed method has considerable promise for use in a VLSI-based system to provide effective cardiac monitoring devices.

CONFLICT OF INTEREST

The authors declare no conflict of interests.

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CONSENT FOR PUBLICATION

Not applicable.

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